

MOHSEN MOSTAFA

Independent Researcher — Physical & Geometric AI

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RESEARCH PROFILE

Independent researcher in Physical & Geometric AI (2023–present), working at the intersection of geometric deep learning, Bayesian and uncertainty-aware architectures, and physics-informed, curvature-aware optimization. Author of 8 first-author preprints produced in under 10 weeks of active writing and validated entirely on free-tier cloud GPUs (Kaggle/Colab T4 15GB, P100 16GB), demonstrating an ability to identify, formalize, and complete original theoretical contributions independently and at pace. Current work extends this program toward embodied prediction and world models — Dynamic LeJEPa (Maximum-Entropy Representations for Sequential Prediction), building on Balestrieri & LeCun's LeJEPa framework — alongside a second line of work on graph-grounded safe routing for adaptive world models. Seeking a doctoral research position to develop this program under formal supervision, with 8+ years of applied AI/ML/CV/GenAI leadership providing the engineering depth to design, implement, and rigorously validate large-scale experiments.

EDUCATION

B.Sc., Management Information Systems

High Institute for Specialized Technological Studies, Cairo, Egypt — 2000–2005

Research training beyond the formal degree: Research Methodology (University of London) and Total Data Quality (University of Michigan) — see Certifications — completed alongside the independent publication record above as preparation for doctoral-level research.

PUBLICATIONS & PREPRINTS

Self-directed research program defining "Physical & Geometric AI." All entries below are preprints (Preprints.org / arXiv), not yet peer-reviewed unless noted; full and updated list maintained on ORCID.

1. Mostafa, M. (2026). Geometric Foundations of Racing Dynamics: How Gradient Descent Adapts Network Capacity on Data Manifolds. Preprints.org, 202603.0760/v1.
2. Mostafa, M. (2025). Gaussian-Constrained LeJEPa Representations for Unsupervised Scene Discovery. arXiv:2602.07016.
3. Mostafa, M. (2026). HPST: A Hybrid Physics-Spectral-Threshold Framework for Fluid Flow Analysis. Preprints.org, 202603.1281/v1.
4. Mostafa, M. (2025). R-LayerNorm: Robust Layer Normalization with Adaptive Noise Suppression. Preprints.org, 202603.0060.
5. Mostafa, M. (2026). Bayesian R-LayerNorm: Uncertainty-Aware Adaptive Normalization with Provable Robustness Bounds. Preprints.org, 202603.0740/v1.
6. Mostafa, M. (2026). Bayesian PASA: Probabilistic Adaptive Sigmoidal Activation with Uncertainty Quantification. Preprints.org.
7. Mostafa, M. (2026). Uncertainty-Aware Classifier with Physics-Based Rejection (UA-PBR). Preprints.org, 202603.0748/v1. [**Live Interactive Demo** : <https://huggingface.co/spaces/MohsenAI5/UA-PBR-Demo>]
8. Mostafa, M. (2026). EPANG-Gen: A Curvature-Aware Optimizer with Uncertainty Quantification for Scientific ML. Preprints.org, 202603.0692/v1.
9. Mostafa, M. (in progress). Dynamic LeJEPa — Maximum-Entropy Representations for Sequential Prediction with Theoretical Guarantees.

PROFESSIONAL EXPERIENCE

AI Scientist & Lead — GraviLog (Remote — US/UAE) (Sep 2025 – Present)

- Designed end-to-end multimodal AI system architecture (CV + LLM) with TensorRT/ONNX optimizations, achieving a 40% latency reduction.
- Defined the technical roadmap for 3 products and delivered strategic research presentations to C-level stakeholders.
- Led the RAG retrieval pipeline (Pinecone + embeddings); mentored 4 engineers in MLOps and system design.

Senior AI Developer (Research Focus) — OMC Technologies (Remote — Limerick, Ireland) *(May 2023 – Aug 2025)*

- Built AI agents integrating LLMs (Falcon, DeepSeek, Qwen, Llama), automating 3 workflows and cutting manual effort by 40%.
- Deployed a production RAG system (LangChain + Pinecone) that reduced hallucinations by 35%.
- Shipped a multimodal healthcare AI system: F1 +19%, latency reduced from 210ms to 145ms via streaming + FP16.
- Built a fraud detection system: AUC +17%, false positives -22%; FastAPI service at p99 ≤ 95ms @ 500 RPS.
- Architected an InsurTech PoC for automated roof damage assessment, fine-tuning SAM 3 via Roboflow on drone imagery, with a custom mIoU-to-area and MAPE-driven cost estimation pipeline.
- Managed AI infrastructure spanning 24 GPUs (12 nodes) and 15 CPU nodes for real-time IoT anomaly detection; optimized distributed training/inference with TensorRT, Kubeflow, and KServe.
- Mentored 5 junior engineers, leading to 2 promotions.

AI Developer & Researcher — Nexgro (South Africa) *(Apr 2020 – Apr 2023)*

- Built a grain quality computer vision pipeline achieving 98% accuracy across 6 grain types.
- Delivered drone/satellite anomaly detection (NDVI + YOLOv8), raising manual inspection efficiency by 160%.
- Optimized 3D point cloud segmentation for LiDAR/DSM data, reaching sub-300ms inference on edge devices.
- Deployed an LLM RAG assistant over technical manuals, saving 15 person-hours per week.

Data Scientist — Nexgro (South Africa) *(Mar 2018 – Mar 2020)*

- Built forecast models that reduced logistics costs by 18%.
- Predictive maintenance models prevented 3 breakdowns, saving roughly \$45K.
- Inventory optimization reduced spoilage by 18%.

Systems Administrator — Egyptian Co. *(Jun 2013 – Mar 2018)*

- Managed IT infrastructure for 200+ users, sustaining 99.8% uptime across multi-site networks and servers.
- Automation scripts cut manual ticket handling by 25%; self-directed pivot into data science (certifications 2016–2018).

IT Network Administrator / Specialist — International Co. for Trade *(Aug 2005 – Jun 2013)*

- Administered LAN/WAN, firewalls, and backup solutions for 150+ nodes; first-line network support and vendor coordination.

TECHNICAL SKILLS

- Deep Learning: PyTorch, TensorFlow, Hugging Face (Falcon, Llama, Mistral, Qwen), JEPa, CLIP, YOLO, Mask2Former
- Probabilistic ML: Bayesian deep learning, uncertainty quantification, natural gradient methods, curvature-aware optimization
- GenAI / LLM: RAG, ColBERT, LangChain, DeepSeek, RLHF concepts, streaming inference
- Computer Vision & Robotics: OpenCV, 3D point cloud segmentation (LiDAR/DSM), multi-view geometry, ROS2, software robotics
- MLOps & DevOps: MLflow, Kubeflow, Airflow, Docker, Kubernetes, KServe, FastAPI, GitLab CI, GCP Vertex AI, AWS SageMaker
- Streaming & Big Data: Apache Kafka, Spark Structured Streaming, Hadoop HDFS, Parquet
- Optimization & Serving: TensorRT, ONNX, model pruning/quantization, FP16 inference
- Data & Databases: Pandas, Dask, Polars, PySpark, SQL (PostgreSQL), MongoDB, Redis, Pinecone

CERTIFICATIONS & RESEARCH TRAINING

- Research Methodology — University of London
- Total Data Quality — University of Michigan
- Deep Learning Specialization — DeepLearning.AI
- TensorFlow Developer & Advanced TensorFlow — Google
- Google Cloud MLOps — Google
- IBM Data Science Professional Certificate — IBM
- CCNA 200-301 & ENCORE 350-401 — Cisco

LEADERSHIP & LANGUAGES

- Mentored 5+ engineers across OMC Technologies and GraviLog, resulting in 2 direct promotions.
- Conducted internal LLM/MLOps workshops and knowledge-transfer sessions.
- Designed a full MLOps showcase pipeline: Kafka → Spark Streaming → HDFS → MongoDB → PostgreSQL → MLflow → KubeFlow → FastAPI for real-time predictive maintenance.
- Languages: Arabic (native), English (fluent, C1/C2).